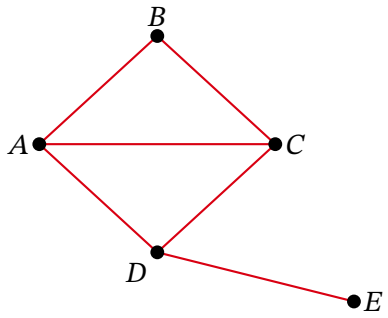
Course notes

Vocabulary

Definition · Graph. A (undirected) *graph* consists of a finite set of *vertices* and a set of *edges*, each edge joining two distinct vertices; two joined vertices are *adjacent*. We consider graphs without multiple edges. The *order* of the graph is its number of vertices. The *degree* of a vertex is the number of edges of which it is an end. A graph is *complete* if any two distinct vertices are adjacent.

Definition · Walk, cycle, connectedness. A *walk* is a sequence of vertices in which any two consecutive vertices are adjacent; its *length* is its number of edges (a vertex or an edge may be used again). A walk is *closed* if its ends are equal. A *cycle* is a closed walk of length at least 3 whose edges are distinct. A graph is *connected* if any two vertices are joined by a walk.



The network of the activity: a graph of order 5 with 6 edges.

Property · Sum of the degrees. In a graph, the sum of the degrees of the vertices is twice the number of edges. Consequently, the number of vertices of odd degree is even.

Proof. Each edge has two ends: it is counted twice in the sum of the degrees. The sum of the degrees is even; now a sum of integers is even if and only if it contains an even number of odd terms. □

Property · Complete graph. The complete graph of order n has $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.

Definition · Directed graph, weighted graph. In a *directed* graph, the links, called *arcs*, have a direction: the arc (X, Y) goes from X to Y. A *path* is a sequence of vertices in which each is joined to the next by an arc. A graph is *weighted* when each edge or arc carries a number, called its weight.

Adjacency matrix

Definition · Adjacency matrix. The vertices of a graph of order n being numbered from 1 to n, its *adjacency matrix* is the square matrix M of order n whose entry m_{ij} is 1 if the vertices i and j are joined (by an edge, or by an arc from i to j for a directed graph), and 0 otherwise. For an undirected graph, M is symmetric: $m_{ij} = m_{ji}$.

Property · Number of walks of length n. Let M be the adjacency matrix of a graph and $n \geq 1$ an integer. The entry in row i and column j of M^n equals the number of walks (of paths, for a directed graph) of length n from vertex i to vertex j.

Proof. Let $m_{ij}^{(n)}$ be the entries of M^n and $c_{ij}^{(n)}$ the number of walks of length n from i to j. For $n = 1$, a walk of length 1 from i to j is an edge: $c_{ij}^{(1)} = m_{ij}$.

Assume $m_{ij}^{(n)} = c_{ij}^{(n)}$ for all i, j. A walk of length $n+1$ from i to j consists of a walk of length n from i to a vertex k, followed by an edge from k to j. Sorting these walks by the second-to-last vertex k, we obtain

$$c_{ij}^{(n+1)} = \sum_{k=1}^{p} c_{ik}^{(n)} m_{kj} = \sum_{k=1}^{p} m_{ik}^{(n)} m_{kj} = m_{ij}^{(n+1)},$$

where p is the order of the graph, since $M^{n+1} = M^n M$. The property holds for $n+1$. □

Key point · Consequences. The diagonal entry $\left(M^2\right)_{ii}$ is the degree of vertex i (undirected graph).

Beyond the syllabus · Connectedness and powers of the adjacency matrix. A graph of order $p \geq 2$ is connected if and only if every off-diagonal entry of $M + M^2 + \cdots + M^{p-1}$ is strictly positive. Indeed, by the theorem, the entry with indices $i \neq j$ counts the walks from i to j of length 1 to $p-1$. If it is strictly positive, i and j are linked. Conversely, if i and j are linked, choose a walk from i to j of minimal length. It never passes twice through the same vertex: otherwise, removing the loop between the two visits would give a shorter walk. So it visits at most p distinct vertices and its length is at most $p-1$. This criterion is not part of the syllabus and is not assessed.